

Parameter Optimisation and Performance Benchmarking of Quantum Annealing for N-Queens

L. Tatli

Level 3 Computing Project, Department of Physics, Durham University

Submitted: March 30, 2025

We investigate the application of a Path Integral Monte Carlo quantum annealing (PIQA) algorithm to the N-Queens problem ($N = 8$). We explore how key parameters, such as the Trotter number P , Trotter temperature PT , and initial transverse field strength Γ_0 , affect convergence and performance. Benchmarking against Monte Carlo simulated annealing (SA), we find that PIQA outperforms SA in finding valid solutions, especially at larger annealing durations of $\tau > 10^3$. For $\tau = 800$ and $PT = 1.6$, we observe near-perfect convergence averaged over 40 runs when $P \geq 16$ and $\Gamma_0 \geq 80$. This improvement becomes significant when P is large enough to suppress systematic Trotter error. We verify differences in how SA and PIQA explore the energy landscape of N-Queens, providing a proof of principle of their underlying physical mechanisms. These results support the viability of quantum-inspired algorithms, as the observed behaviour is likely to generalise to a broader class of combinatorial optimisation problems. Future work could extend to larger instances and to implementation on real quantum annealers.

I. INTRODUCTION

Optimisation problems are widely recognised as being NP-hard, meaning that while verifying a proposed solution can be computationally manageable, finding a solution efficiently remains an outstanding challenge. Many real-world applications, ranging from timetable scheduling to resource allocation, fall under the umbrella of combinatorial optimisation (CO) problems, where the objective is to identify the global minimum of a discrete cost (or energy) function defined over a large solution space [1]. In such landscapes, the presence of numerous local minima often causes conventional search methods to become trapped, thus motivating the development of heuristic approaches.

Among these heuristics, simulated annealing (SA) and quantum annealing (QA) have emerged as useful techniques. Both methods draw inspiration from natural physical processes: SA mimics a thermal cooling process in metallurgy, wherein thermal fluctuations allow the system to escape local minima, while QA uses quantum tunnelling, mediated by a transverse field term Γ , to traverse energy barriers.

SA, as introduced by Kirkpatrick et al. [2], has been successfully applied to a wide range of CO problems. However, recent studies have provided evidence suggesting a "quantum advantage" of QA over SA in certain contexts. For example, investigations into finding the ground state of Ising ferromagnets [3] and comparative studies on two-dimensional spin-glass Ising models [4] indicate that QA can outperform SA. These findings were further shown by Martoňák et al. [5] when comparing the performance of SA with Path Integral Monte Carlo (PIMC) QA, in a numerical technique known as Path Integral Quantum Annealing (PIQA), in the Travelling Salesman Problem (TSP), a popular CO problem. Nevertheless, the literature on PIQA remains sparse and its performance appears to be problem-dependent; for example, in the case of 3-SAT, no quantum advantage was observed [6].

In this work, we investigate the application of PIQA to the N-Queens puzzle, in which N queens must be placed on an $N \times N$ chessboard such that no two queens threaten each other. Originally proposed in the 1800s, the N-Queens problem is a well-known NP-hard problem. The choice of N-Queens as the focus of this study was not arbitrary; this problem has found various useful applications, including neural networks and traffic control [7]. Moreover, its unique energy landscape, as detailed in Section II, make it

a compelling yet relatively tractable problem to encode.

Although *ad-hoc* approaches such as backtracking can efficiently find solutions, our focus is not on developing a specialised solver but rather on comparing the performance of PIQA with that of SA, and on determining the optimal parameter regime for PIQA. Our implementation follows the general methodology outlined by Martoňák et al., by mapping the problem to an Ising model and applying Trotter decomposition to simulate quantum effects on a classical machine. In addition, we discuss the implications of our results and draw comparisons with other approaches to solving CO problems, suggesting avenues for future research.

II. ENCODING THE N-QUEENS PROBLEM

We begin by translating the problem constraints into a mathematical framework. For the classical N-Queens problem, a valid solution requires that each row and each column contain exactly one queen, and that no two queens share a diagonal. We represent an $N \times N$ chessboard as a matrix S with elements S_{ij} (for $0 \leq i, j \leq N - 1$), representing Ising spin variables that take values in $\{-1, +1\}$. A queen is present at the position (i, j) if $S_{ij} = +1$ and absent if $S_{ij} = -1$. Following the approach of Tsukiyama et al. [8], we convert these spins into binary indicator variables via

$$x_{ij} = \frac{1 + S_{ij}}{2}, \quad (1)$$

where

$$x_{ij} = \begin{cases} 1, & \text{if a queen is present at } (i, j) \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

This conversion facilitates the expression of constraints in a more intuitive way of visualising our problem, i.e. a Quantum Unconstrained Binary Optimisation (QUBO) formulation, that is mathematically equivalent to the Ising model. In this formulation, a "one-hot" property is essential. For instance, a one-hot vector representing a row of the matrix must have exactly one entry equal to 1 (and the remaining $N - 1$ entries equal to 0), which is equivalent to the condition

$$\sum_{j=0}^{N-1} x_{ij} = 1. \quad (3)$$

We enforce this row constraint by including a quadratic penalty term and 0-based indexing

$$H_{\text{row}} = A \sum_{i=0}^{N-1} \left(\sum_{j=0}^{N-1} x_{ij} - 1 \right)^2, \quad (4)$$

where $A > 0$ is a scaling constant chosen to strongly penalise any deviation from one queen per row; physically, this increases the height of the energy barriers. An analogous expression,

$$H_{\text{col}} = B \sum_{j=0}^{N-1} \left(\sum_{i=0}^{N-1} x_{ij} - 1 \right)^2, \quad (5)$$

with $B > 0$, enforces the column constraint. Finally, we require that diagonals (or anti-diagonals) contain at most one queen (i.e., zero or one). For the main diagonals, identified by the index $d = i - j$ (where $-N + 1 \leq d \leq N - 1$), we impose zero-one-hot constraints

$$H_{\text{diag}} = C \sum_{d=-N+1}^{N-1} \left(\sum_{i-j=d} x_{ij} \right) \left(\sum_{i-j=d} x_{ij} - 1 \right). \quad (6)$$

To derive an identical penalty H_{adiag} for the anti-diagonals, we replace the index d with $d' = i + j$, where $0 \leq d' \leq 2(N - 1)$, and $C > 0$. The overall QUBO Hamiltonian, which serves as the potential energy function in our optimisation scheme, is given by

$$H_{\text{pot}}(S) = H_{\text{row}} + H_{\text{column}} + H_{\text{diag}} + H_{\text{adiag}}, \quad (7)$$

where a configuration with $H_{\text{pot}}(S) = 0$ corresponds to a valid N-Queens solution.

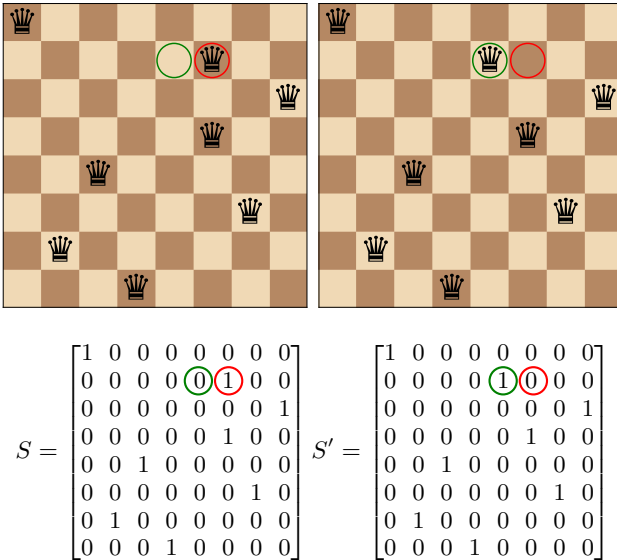


FIG. 1: Top left: An invalid N-queens configuration. Top right: A valid solution. Bottom: The corresponding matrices, with S on the left and the solution S' on the right. Red circles indicate the incorrect position of a queen, while green circles mark the new correct position after a local swap move is implemented.

In our implementation, the exploration of a solution space is achieved through a simple local move known as the swap

move. This transforms the matrix S to a new matrix S' by randomly selecting a cell containing a queen and another cell that is empty, and exchanging their spin values. This operation preserves the total number of queens, ensuring that the one-hot constraints for rows and columns are maintained in aggregate, while allowing the system to escape from local minima in the energy landscape defined by Eq. (7). Fig. (1) displays an example of this move, which leaves all other queens unchanged. The quantum swap move implementation is further explored in Section IV.

III. SIMULATED ANNEALING

In SA, the system begins in an arbitrary initial state S_0 with corresponding energy $H_{\text{pot}}(S_0)$ and explores a configuration space $\{S_T\}$ via iterative modifications; in our case, this modification is implemented by the swap move. At each iteration, a candidate configuration S' is generated and its energy is calculated according to Eq. (7). The change in energy is defined as $\Delta H = H_{\text{pot}}(S') - H_{\text{pot}}(S)$, and the candidate configuration is accepted with probability given by the Metropolis criterion

$$P_{\text{accept}} = \min \left(1, \exp \left(-\frac{\Delta H}{T_{\text{SA}}} \right) \right), \quad (8)$$

where T_{SA} is the current temperature of the system according to an annealing schedule. In this work, a linear annealing schedule is employed, allowing T_{SA} to decrease uniformly over the course of the simulation.

At each temperature level, a fixed number of Monte Carlo (MC) steps, denoted by τ , are performed. During each MC step, a swap move is attempted. A uniformly distributed random number u in the interval $[0, 1]$ is generated, and the candidate move is accepted if $u < P_{\text{accept}}$; otherwise, the move is rejected and the current configuration is retained. This probabilistic acceptance mechanism allows the system to occasionally accept moves that increase the energy, enabling escapes from local minima, particularly at higher temperatures.

As T_{SA} decreases, the probability of accepting moves that result in a higher $H_{\text{pot}}(S)$ diminishes, and the system increasingly favours moves that lower the energy. Physically, this implies that SA is capable of surmounting large potential barriers during the early stages of the process, while at lower temperatures, the system becomes more selective, effectively "freezing" into low-energy states. Over many τ steps, the system is guided toward a configuration that minimises the energy.

Ideally, as $T_{\text{SA}} \rightarrow 0$ and for sufficiently small problem sizes (a small N in our work), the system converges to a state where $H_{\text{pot}}(S) = 0$. For larger problems, where the size of $\{S_T\}$ grows with N , it may be necessary to accept a solution that is "close enough" to the global minimum, which is a common compromise in meta-heuristic optimisation.

IV. QUANTUM ANNEALING

The dynamics of QA are driven by quantum, rather than thermal, fluctuations. In our approach, the new Hamiltonian is written as

$$H(\tau) = H_{\text{pot}}(S) + H_{\text{kin}}, \quad (9)$$

where H_{kin} is an artificial kinetic energy term that enables the system to explore different configurations via a tunnelling mechanism that allows it to avoid becoming trapped in local minima [9].

Following the methodology of Martoňák et al., we note that the N-Queens problem, like the TSP, does not possess a natural H_{kin} term [5]. Instead, one must construct H_{kin} so that it defines valid transitions between configurations, thereby influencing the energy landscape. In this work, we base H_{kin} on the swap move from our SA approach. This move preserves the global constraint that exactly N queens remain on the board at all times, corresponding to a conserved total magnetisation in the Ising model. We write

$$H_{\text{kin}} = -\Gamma(\tau) \sum_{(i,j)} \sum_{(k,l)} S_{(k,l)}^+ S_{(i,j)}^-, \quad (10)$$

where $(i,j) \in Q$ is a site containing a queen, $(k,l) \notin Q$ is a site without a queen, and $S_{(k,l)}^+$ and $S_{(i,j)}^-$ are spin-raising and spin-lowering operators, respectively.

Physically, $\Gamma(\tau)$ measures the strength of quantum tunnelling between states. Starting from a moderate initial value Γ_0 , we gradually reduce $\Gamma(\tau)$ via a linear schedule, similar to the T schedule in SA. When $\Gamma(\tau)$ is large, H_{kin} dominates, allowing the system to tunnel through high but narrow barriers in the energy landscape. As $\Gamma(\tau)$ is reduced, the system is allowed to settle into a configuration near or at the global minimum.

In practice, such a scheme can be implemented on specialised quantum annealers [10], although solving large CO problems is computationally intensive, and quantum systems are challenging to simulate directly. Consequently, we adopt PIQA, which is a numerical implementation on classical hardware to emulate QA.

V. PATH INTEGRAL MONTE CARLO QUANTUM ANNEALING

PIQA is a stochastic method employed to simulate the equilibrium properties of a quantum system at a fixed, finite temperature T . PIQA approximates QA by mapping the quantum partition function of Eq. (9) to an equivalent classical partition function via a Trotter decomposition. This transformation replaces the original d -dimensional Ising model with an equivalent classical model in $d+1$ dimensions. In practice, we construct a stack of P classical replicas (or Trotter slices), coupled along a new imaginary-time (Trotter) dimension. The resulting classical Hamiltonian is expressed as

$$H_{d+1} = \sum_{k=1}^P \left(H_{\text{pot}} + J^\perp \sum_{(i,j)} S_{(i,j)}^k S_{(i,j)}^{k+1} \right), \quad (11)$$

where $S_{(i,j)}^k$ represents the classical spin variable at site (i,j) in the k th Trotter slice. The first term represents H_{pot} for each k , and the second term couples adjacent slices along the Trotter dimension. The coupling strength is given by

$$J^\perp = -\frac{PT}{2} \ln \left(\tanh \frac{\Gamma}{PT} \right), \quad (12)$$

where T is the physical temperature of the quantum system, and PT is the Trotter temperature of the equivalent classical

system. A full derivation of this Trotter discretisation is provided in Refs. [11] and [12], though we outline the main details in Appendix A. Fig. (2) provides a visualisation of the Trotter slices for clarity.

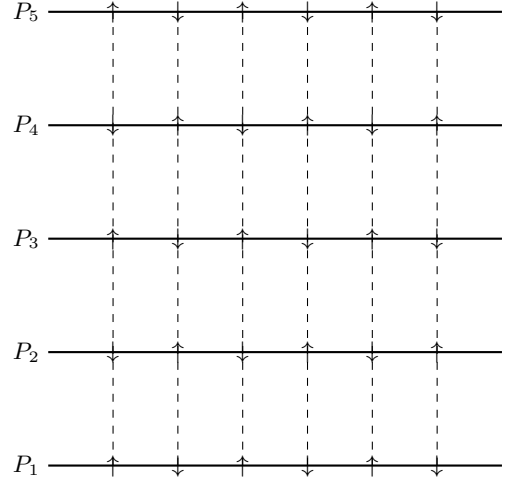


FIG. 2: Trotter slices with spin states at each node. Each horizontal line represents a Trotter slice P in the path integral formulation, with nodes showing the spin state (up or down). Dashed vertical lines indicate couplings between the slices.

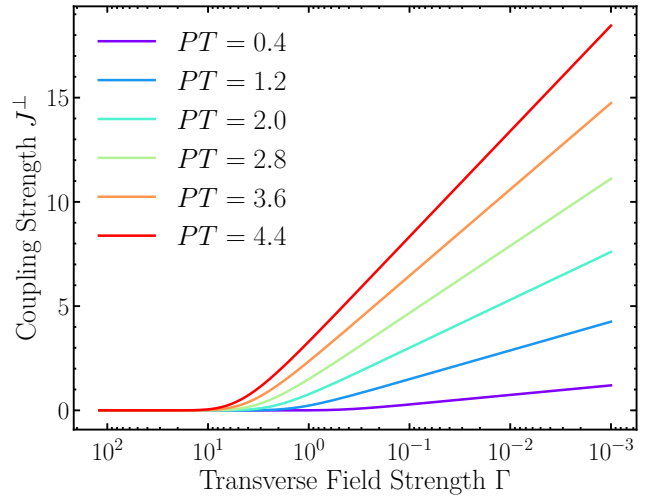


FIG. 3: Plotting Eq. (12), the coupling strength $J^\perp$ between adjacent Trotter slices as a function of the transverse field strength Γ . Each line represents a different Trotter temperature PT .

The accuracy of this numerical approach is governed by the Trotter number P . In the limit $P \rightarrow \infty$, the classical system restores the original quantum system. However, computational constraints require P to remain finite, introducing a systematic approximation known as the Trotter error. This error diminishes as $O(\frac{1}{P^2})$, consequently, larger P values permit more accurate quantum simulations, albeit at the cost of greater computational effort due to the larger configuration space. Details of this systematic error analysis are provided in Appendix B.

Practically, the PIQA simulation proceeds as follows. Initially, when $\Gamma(\tau)$ is large, the coupling $J^\perp$ is very weak, and the P Trotter slices are nearly independent at temperature PT . To ensure equilibrium, we initialise each slice to

an identical random configuration, generating P copies of a random chessboard. A plot of $J^\perp$ as a function of Γ for different PT is shown in Fig. (3).

Once initialised, the annealing schedule reduces $\Gamma(\tau)$ linearly from Γ_0 towards a small final value close to zero. Selecting an appropriate Γ_0 is critical; we investigated a range of values and provide a discussion in Section VII. At each MC step, the system is sampled using Eq. (8) and swap moves are applied to each of the P Trotter slices. The candidate energy of the updated configuration is evaluated using Eq. (11). The temperature T is maintained constant throughout the τ steps, and upon completion, the Trotter slice with the lowest energy is identified; its energy is recorded using Eq. (7) as the final result. It should be noted that PIQA inherently demands more computational resources than SA, specifically scaling with P . Hence, while potentially superior in performance, PIQA is typically a factor of P slower per MC step τ than its classical counterpart.

VI. NUMERICAL IMPLEMENTATION

All simulations of SA and PIQA were performed in Python using the `numpy` library for scientific operations and random sampling. We selected $N = 8$ (i.e. 8-Queens) as the focus of this study due to its relative ease of implementation and straightforward identification of solutions. Larger N values were not investigated. This allowed us to verify by inspection that the simulations converged to one of the 92 distinct solutions of 8-Queens [13]. In H_{pot} , scaling constants $A, B, C = 2$ were found to be sufficiently effective in enforcing the queen restrictions whilst still permitting tunnelling to occur, and in maintaining potential barriers at a level that allowed SA to perform.

The code is available in Ref. [14]. Within `nqueens.py`, the functions `RunSA` and `RunPIQA` were developed to return both the final energy and the optimal chessboard in a readable format. Due to finite sampling, both approaches are subject to random error; therefore, the general performance of each algorithm was captured by running several independent simulations with different random seeds and averaging the results. Details of these error calculations are provided in Appendix B.

Determining the optimal ranges for the parameters Γ_0 , P , and PT was a non-trivial but important task. As these parameters are specific to the problem being encoded, it was challenging to reproduce the values from the literature. We resorted to trial and error, performing convergence tests for each parameter whilst keeping the others constant. To facilitate this process, particularly for multiple averaging to ensure reliable results, we employed parallel computing using the `multiprocessing` library.

VII. RESULTS AND DISCUSSION

Since $H_{\text{pot}} = 0$ corresponds to a valid solution, we assess the effectiveness of SA and PIQA by comparing their absolute final energies averaged over several runs, ϵ_{final} . In what follows, we first discuss the dependence of PIQA on Γ_0 , P , and PT . We then present a comparative analysis of PIQA versus SA as a function of annealing time τ .

A. Parameter Dependence of PIQA

We chose $\tau = 800$ as the suitable annealing duration for all runs in this section. As highlighted in Section V and in Fig. (3), the Trotter temperature PT sets both the inter-slice coupling strength $J^\perp$ and the effective thermal fluctuations between spins in each Trotter slice. Meanwhile, the Trotter number P controls how finely the imaginary-time dimension is discretised. By examining these two parameters independently, we can see more clearly how each contributes to the average final energy ϵ_{final} .

Fig. (4) shows ϵ_{final} as a function of the initial transverse field strength Γ_0 and P for several fixed PT values between 0.40 and 4.40. We find that at the lower end of this range, around $PT = 0.40$, $J^\perp$ is too weak to sustain tunnelling when Γ_0 is large, leading the system to “freeze” into high-energy states before the end of the anneal. In effect, when P is too low, the discretisation is too coarse to accurately represent quantum effects, causing the system to resemble a classical annealer and lose its tunnelling advantage prematurely. As we increase PT , the replica slices remain correlated for longer, resulting in better overall convergence, even though an excessively low Γ_0 (below approximately 60) does not allow the annealing period to be sufficiently long to achieve convergence. By contrast, a large Γ_0 does not negatively impact convergence provided that the Trotter discretisation is sufficiently fine, suggesting that a strong $J^\perp$ can enhance performance when adequately resolved.

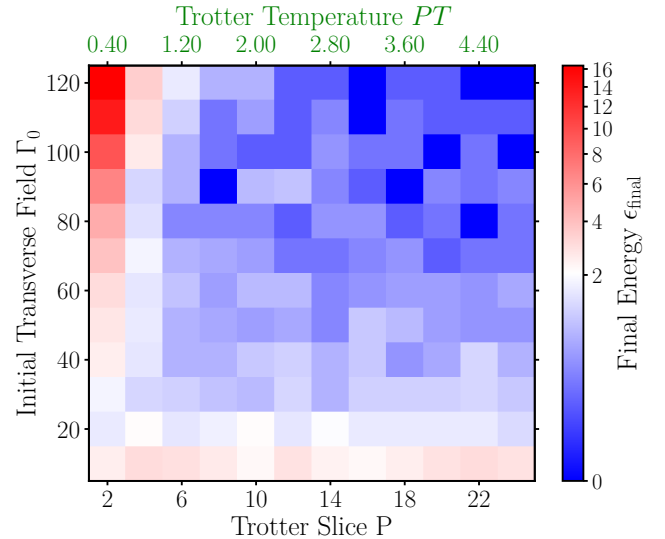


FIG. 4: Heatmap showing the relationship between the initial transverse field strength Γ_0 and Trotter number P for PIQA. The colour scale indicates the final energy ϵ_{final} averaged over 40 independent runs. The upper axis (in green) displays the corresponding PT values. All simulations were performed with $\tau = 800$, annealed to $\Gamma_{\text{final}} = 0.001$, using $T = 0.02$.

These results mirror Battaglia et al.’s description of PIQA [6]. In the early phase, there is a quenching period during which Γ dominates while $J^\perp$ is too small to effectively couple the replicas, causing each Trotter slice to behave like an independent MC simulation. Next, in the search phase, $J^\perp$ increases, allowing tunnelling between energy valleys, provided that P is sufficiently high. Finally, as $\Gamma \rightarrow 0$, the replicas enter a selection or freezing phase in which strong

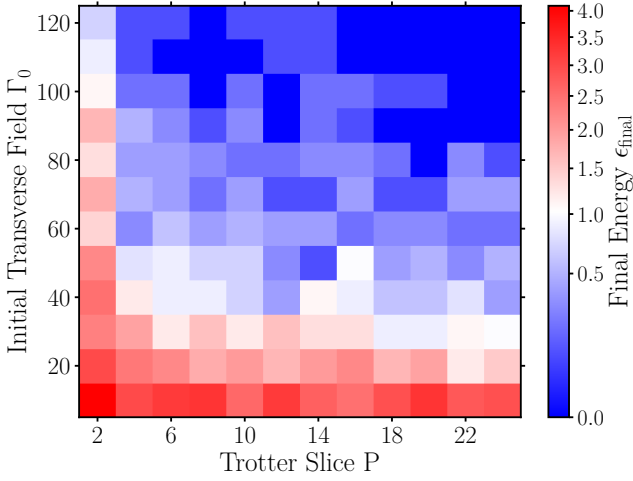


FIG. 5: Heatmap showing the relationship between the initial transverse field strength Γ_0 and Trotter number P for PIQA at $PT = 1.6$. The colour scale indicates the final energy ϵ_{final} averaged over 40 independent runs. All simulations were performed with $\tau = 800$, annealed to $\Gamma_{\text{final}} = 0.001$, using a scaled T for each P to maintain a constant PT .

coupling forces them into a single classical configuration and the quantum effects are “switched off”. If P is too small, the final convergence can occur too early, trapping the system in a suboptimal configuration (“classical freezing”), as seen in the red regions of Fig. (4) in the top left. If P is large enough, the tunnelling continues until later in the schedule (“quantum freezing”), often yielding lower energies, as shown by the dark blue regions in the top right. Clearly, this latter parameter setup is preferred for PIQA.

A more detailed study of $PT = 1.6$ is presented in Fig. (5), focusing on this intermediate value to balance computational cost with improved performance. We find that the worst convergence occurs at $P = 2$, where the quantum system is not approximated sufficiently well. Increasing P at a constant PT significantly reduces ϵ_{final} , especially when Γ_0 is large enough to ensure the system spends more time in the search phase, permitting tunnelling. Conversely, when Γ_0 is very low (below about 40), the system shows signs of partial classical freezing. In this regime, even a large P struggles to overcome shallow energy barriers, as insufficient quantum fluctuations do not provide the necessary tunnelling advantage. Although Martoňák et al. propose that further increases in P eventually hit diminishing returns [4], our tests do not show a clear plateau. We find that the region of optimal performance, where the algorithm consistently converges to near-perfect ($\epsilon_{\text{final}} \lesssim 0.5$) or perfect solutions, is located in the dark blue area in the top right of Fig. (5), corresponding approximately to $\Gamma_0 \geq 80$ and $P \geq 16$.

Because heatmaps hide the random error across multiple runs, we examined a cross-section at $\Gamma_0 = 60$ and plotted ϵ_{final} versus P , as seen in Fig. (6). At low P , PIQA behaves similarly to SA, and the standard error is relatively large. As demonstrated in Appendix B, the systematic error due to the Trotter approximation is negligible for $P \geq 5$. Beyond $P \approx 14$, both the average energy and its error decrease considerably, indicating a clear improvement in tunnelling capability. This improvement eventually levels off, suggesting smaller gains for each additional Trotter slice. This

trade-off between deeper quantum sampling (i.e., larger P) and increased computational cost could be attributed to our implementation, which currently only attempts local swap moves. Introducing global updates, such as by flipping spins in all Trotter replicas at the same site, might alter the dynamics and lead to faster convergence, although this has not been explored in detail.

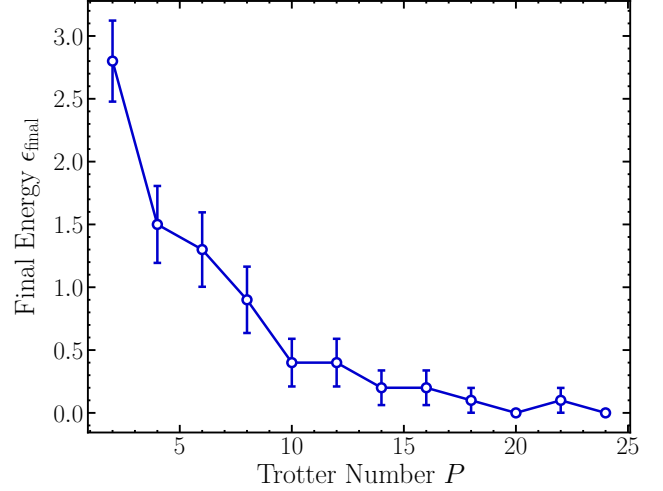


FIG. 6: Final energy averaged over 40 runs against the Trotter number P for PIQA at $PT = 1.6$. All simulations were performed with $\tau = 800$, annealed from $\Gamma = 60$ to $\Gamma_{\text{final}} = 0.001$, using a scaled T for each P to maintain a constant PT . Error bars represent the standard error of the mean.

B. Comparison between SA and PIQA

Fig. (7) shows a comparison between SA and PIQA as a function of τ . For a fair comparison, we calculated the normalised final energy $\tilde{\epsilon}_{\text{final}}$, and account for the fact that PIQA’s computational cost scales linearly with P ; therefore, we plot the PIQA results as a function of $\tau \times P$. This allows us to directly compare the effective CPU time or number of MC updates across the two methods. All SA runs were annealed from a sufficiently high temperature $T_0 = 100$.

Our results indicate that PIQA outperforms SA, especially at $\tau > 10^3$. In this regime, PIQA achieved a 100% success rate, with $\tilde{\epsilon}_{\text{final}} = 0$ across all independent runs. In contrast, SA converged to $\tilde{\epsilon}_{\text{final}} \approx 0.2$, with no runs achieving a perfect solution. Moreover, while the standard error in SA appears to decrease over time, we did not observe a similar reduction in the error for PIQA.

As outlined in Section II, the scaling constants A , B , and C were each set to 2 in our study, significantly influencing the energy landscape. Since these constants determine the penalty for constraint violations, they inherently affect how each algorithm performs. QA has been shown to benefit from tall, narrow energy barriers that enable Landau-Zener tunnelling. In contrast, SA performs better when the energy barriers are lower, as this enables the algorithm to “climb” over local minima more efficiently. Santoro et al. attributed the observed quantum advantage to cascades of Landau-Zener tunnelling events, which allow the system to traverse multiple energy barriers that would otherwise slow down the performance of SA [15].

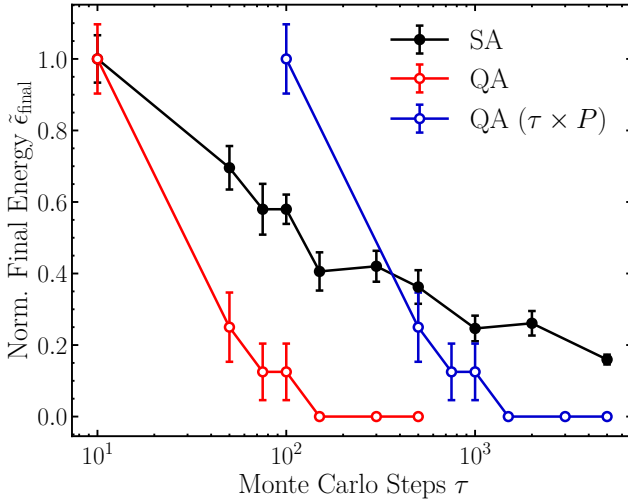


FIG. 7: Normalised final energy $\tilde{\epsilon}_{\text{final}}$, averaged over 5 runs, plotted against the number of Monte Carlo steps τ for SA (black), PIQA (red), and PIQA scaled by P (blue). All PIQA runs were conducted with $P = 10$ and $PT = 1$, annealed from $\Gamma_0 = 50$ to $\Gamma_{\text{final}} = 0.001$. SA runs were annealed from $T_0 = 100$.

VIII. CONCLUSIONS AND OUTLOOK

In this work, we investigated the possibility of solving the N-Queens problem using a numerical QA approach, PIQA, and compared our results to a classical SA approach. Our findings indicate a potential quantum advantage for $N = 8$. We stress that annealing methods are not the optimal algorithm for N-Queens; rather, our study serves as a proof of principle for examining the underlying mechanisms.

The results encourage further exploration of quantum algorithms for solving CO problems more broadly. Existing literature suggests that the effectiveness of such algorithms strongly depends on the specific mapping of the problem onto an energy landscape, perhaps even more so than the intrinsic efficiency of the algorithm itself. Extending this to real quantum annealers represents a natural next step. In their recent work, Tsukiyama et al. outlined a method to reduce the number of terms required for encoding N-Queens by introducing domain wall vectors, thus optimising the resource usage on practical quantum hardware [16]. In our study, we adopted a direct QUBO formulation, as it remains more straightforward to implement on a classical machine.

Further extensions of this research might explore problem sizes larger than $N = 8$, since the primary interest in quantum algorithms lies in addressing computationally challenging scenarios where classical methods struggle. Limitations inherent in our simple PIQA approach also motivate consideration of alternative sampling methods, such as Green's Function Monte Carlo [5].

The broader potential of quantum algorithms continues to be an active and growing area of research. Besides ongoing developments of physical QA machines, recent studies have introduced hybrid algorithms that integrate QA with classical methods, such as the Quantum Approximate Optimisation Algorithm (QAOA) [17]. Variations of PIQA have also been developed, incorporating local search techniques and further enhancing the algorithm's capabilities [18]. Exploring these directions may help to fully realise the advantages offered by quantum approaches in solving real-world

applications of CO problems.

REFERENCES

- [1] C. H. Papadimitriou and K. Steiglitz, *Combinatorial Optimization: Algorithms and Complexity*, Englewood Cliffs, NJ: Prentice-Hall, 1982, ISBN: 0-13-022163-9
- [2] S. Kirkpatrick et al., "Optimization by simulated annealing," *Science*, vol. 220, no. 4598, pp. 671–680, May 1983, doi:10.1126/science.220.4598.671.
- [3] J. Brooke et al., "Quantum Annealing of a Disordered Magnet," *Science*, vol. 284, no. 5415, pp. 779–781, Apr. 30, 1999, doi:10.1126/science.284.5415.779.
- [4] R. Martoňák et al., "Quantum annealing by the path-integral Monte Carlo method: The two-dimensional random Ising model," *Phys. Rev. B*, vol. 66, p. 094203, Sep. 2002, doi:10.1103/PhysRevB.66.094203.
- [5] R. Martoňák et al., "Quantum annealing of the traveling-salesman problem," *Phys. Rev. E*, vol. 70, p. 057701, Nov. 2004, doi:10.1103/PhysRevE.70.057701.
- [6] D. A. Battaglia et al., "Optimization by quantum annealing: Lessons from hard satisfiability problems," *Phys. Rev. E*, vol. 71, p. 066707, Jun. 2005, doi:10.1103/PhysRevE.71.066707.
- [7] J. Bell and B. Stevens, "A survey of known results and research areas for n-queens," *Discrete Mathematics*, vol. 309, pp. 1–31, 2009, doi:10.1016/j.disc.2007.12.043.
- [8] S. Tsukiyama et al., "Solving the N-Queens Puzzle by a QUBO Model with Quadratic Size," in *Proc. Eleventh Int. Symp. on Computing and Networking (CANDAR)*, Matsue, Japan, Nov. 28–Dec. 01, 2023, doi:10.1109/CANDAR60563.2023.00015.
- [9] C. C. McGeoch, *Adiabatic Quantum Computation and Quantum Annealing: Theory and Practice*, Synthesis Lectures on Quantum Computing. Cham, Switzerland: Springer Cham, 2014, ISBN: 978-3-031-01390-4.
- [10] M. W. Johnson et al., "Quantum annealing with manufactured spins," *Nature*, vol. 473, no. 7346, pp. 194–198, 2011, doi:10.1038/nature09912.
- [11] A. Das and B. K. Chakrabarti, *Quantum Annealing and Related Optimization Methods*, Lecture Notes in Physics, vol. 679. Berlin, Heidelberg: Springer, 2005, ISBN: 3-540-27987-3.
- [12] M. Suzuki, "Relationship between d-Dimensional Quantum Spin Systems and (d+1)-Dimensional Ising Systems—Equivalence, Critical Exponents and Systematic Approximants of the Partition Function and Spin Correlations," *Prog. Theor. Phys.*, vol. 56, no. 5, pp. 1454–1469, Nov. 1976, doi:10.1143/PTP.56.1454.
- [13] "The Eight Queens Puzzle," Concordia University, [Online]. Available: <https://users.encs.concordia.ca/~chvatal/8queens.html> (accessed: Mar. 26, 2025).
- [14] L. Tatli, *GitHub repository*, [Online]. Available: <https://github.com/LaraTatli18/quantum-optim-applied-algorithms> (accessed Mar. 26, 2025).
- [15] G. E. Santoro et al., "Theory of quantum annealing of an Ising spin glass," *Science*, vol. 295, no. 5564, pp. 2427–2430, Mar. 2002, doi:10.1126/science.1068774.
- [16] S. Tsukiyama et al., "Ising models for solving the N-queens puzzle based on the domain-wall vectors," *Concurrency and Computation: Practice and Experience*, vol. 37, no. 3, Feb. 2025, doi:10.1002/cpe.8364.
- [17] E. Farhi et al., "A Quantum Approximate Optimization Algorithm," *arXiv:1411.4028 [quant-ph]*, 2014, doi:10.48550/arXiv.1411.4028.
- [18] N. Chancellor, "Modernizing quantum annealing using local searches," *New J. Phys.*, vol. 19, no. 2, 023024, Feb. 2017, doi:10.1088/1367-2630/aa59c4.

APPENDIX A: SUPPLEMENTARY DETAILS OF THE PIQA FORMULATION

We provide a brief outline of the main steps leading from Eq. (9) to Eq. (11). Consider the quantum partition function

$$Z = \text{Tr} \left(e^{-\beta H(t)} \right), \quad (13)$$

where $\beta = 1/k_B T$ and $H(t)$ is given by Eq. (9), encoding the quantum statistics of a chessboard configuration. Applying the Trotter break-up formula, we write

$$e^{-\beta H(t)} \approx \left(e^{-\frac{\beta}{P} H_{\text{kin}}} e^{-\frac{\beta}{P} H_{\text{pot}}} \right)^P + O \left(\frac{1}{P^2} \right), \quad (14)$$

which becomes exact in the limit $P \rightarrow \infty$. Inserting resolutions of the identity between each exponential introduces a set of P Trotter slices $\{s_1, s_2, \dots, s_P\}$. Each s_k represents a classical spin configuration at imaginary-time replica k . Hence, the partition function factorises into a sum over all possible Trotter slice configurations:

$$Z \approx \sum_{\{s_k\}} \prod_{k=1}^P \langle s_k | e^{-\frac{\beta}{P} H_{\text{kin}}} | s_{k+1} \rangle e^{-\frac{\beta}{P} H_{\text{pot}}(s_k)}. \quad (15)$$

In the transverse-field Ising case, the matrix element within Eq. (15) takes a simple Ising-like form after applying the Suzuki-Trotter transformation

$$\langle s_k | e^{-\frac{\beta}{P} H_{\text{kin}}} | s_{k+1} \rangle = \exp \left(\beta J^\perp \sum_i s_k^i s_{k+1}^i \right), \quad (16)$$

with the transverse coupling term $J^\perp$ typically depending on Γ , β , and P . Consequently, the original quantum Hamiltonian is transformed to an effective $(d+1)$ -dimensional classical Hamiltonian, with one extra dimension for imaginary time. The error introduced by a finite P decreases as $O(\frac{1}{P^2})$.

APPENDIX B: SYSTEMATIC AND RANDOM ERRORS

To analyse the systematic error introduced by the Trotter approximation, we plot the results of Fig. (6) scaling with $\frac{1}{P^2}$. This is presented in Fig. (8).

As expected, for small values of $\frac{1}{P^2}$ (large P), the final energy decreases very rapidly and approaches zero. Quantitatively, one can observe that when $\frac{1}{P^2} < 0.05$ (i.e. for $P \gtrsim 5$), the differences in ϵ_{final} between successive values of P are smaller than the error bars. This indicates that the Trotter error is negligible in this regime.

The error bars in Figs. (6) and (7) reflect the standard error of the mean. The standard deviation σ_ϵ was determined using `numpy`, and the standard error α_ϵ was computed as

$$\alpha_\epsilon = \frac{\sigma_\epsilon}{\sqrt{r}}, \quad (17)$$

where r is the number of independent simulation runs.

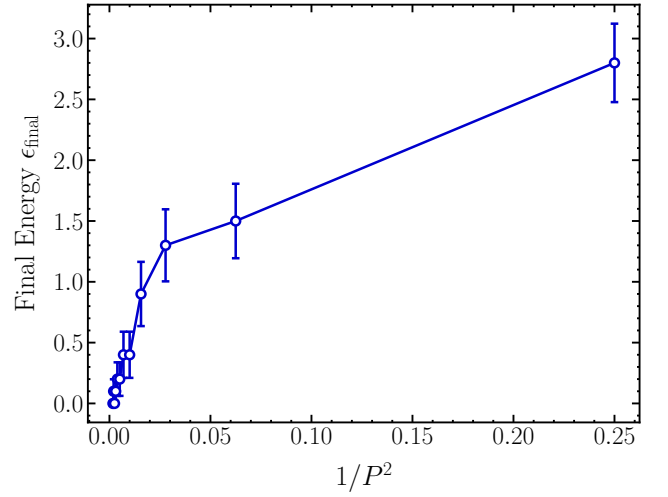


FIG. 8: Average final energy ϵ_{final} versus $\frac{1}{P^2}$ for PIQA. Data points are averaged over 40 independent runs, and the error bars represent the standard error of the mean.

SCIENTIFIC SUMMARY FOR A GENERAL AUDIENCE

The 8-Queens problem, a puzzle enjoyed by chess players, has practical applications beyond the board, including traffic control and satellite communications. The challenge is to place 8 queens on an empty chessboard so that none can attack each other.

One way to solve the problem is to check every possible arrangement. However, this becomes time-consuming as the board size increases. This generalised version, called the N-Queens problem, belongs to a class of problems called combinatorial optimisation.

To address such problems, researchers use physics-inspired techniques known as simulated annealing (SA) and quantum annealing (QA). We can visualise the problem as a mountainous landscape, where the deepest valley represents the best solution. SA randomly explores different paths along the terrain and tends to move downhill, but may become trapped in a high valley. QA introduces a quantum mechanical effect known as quantum tunnelling. Instead of climbing over the hills, it can tunnel through them and find the deepest valley faster than SA.

In this work, we compared SA and QA for solving N-Queens. We found that QA outperforms SA in finding accurate solutions, but at a greater computational cost. Developing quantum-inspired algorithms that improve on current methods remains an active area of research.

GENERATIVE AI STATEMENT

gAI was used to assist with the construction of Fig. (1), and to format equations and references in LaTeX.